

Experiment Name: Study the Role of Power Electronics for Environment and Sustainability

Abstract

Power electronics, the branch of electrical engineering concerned with the conversion and control of electric power using semiconductor switching devices, has become a cornerstone technology for achieving global energy and environmental sustainability. This report studies the role that power electronic converters play in enabling renewable energy integration, smart grid operation, electric transportation, and industrial energy efficiency, thereby reducing greenhouse gas emissions and supporting a cleaner energy future. The discussion combines general concepts with country-specific data for Bangladesh, drawing on figures published by the International Energy Agency (IEA), the Bangladesh Power Development Board (BPDB), and Power Grid Bangladesh PLC (PGCB), to illustrate the present state of the power sector, the extent of renewable energy penetration, and the challenges that remain.

Introduction

Power electronics is the application of electronics to the control and conversion of electric power. Early high-power conversion relied on mercury-arc valves; modern systems instead use semiconductor switching devices such as diodes, thyristors, power MOSFETs, and IGBTs, with variable-speed drive systems ranging in power from a few hundred watts to tens of megawatts. Because almost every stage of modern power generation, transmission, and consumption now passes through some form of power-electronic converter, the technology sits at the center of the transition toward a more sustainable energy system.

Energy sustainability refers to meeting present and future energy demand without exhausting or depleting natural resources, primarily through sources such as solar, wind, geothermal, hydropower, and ocean energy that are generally available at no fuel cost and can be used repeatedly. Environmental sustainability, as defined by the United Nations World Commission on Environment and Development, means acting in a way that ensures future generations retain access to the natural resources needed for a quality of life at least equal to that of the present generation, while improving human welfare within the carrying capacity of the Earth's supporting ecosystems.

Key Features of Power Electronics for Sustainability

Power electronic converters act as the interface that transfers electricity from one stage of the supply chain to the next, and this flexibility is what allows several key sustainability outcomes to be achieved:

1. Renewable energy integration: power converters (rectifiers, inverters, DC-DC converters) are the key technology that allows variable-output sources such as solar photovoltaic and wind turbines to be connected safely to the grid.
2. Smart grid enablement: power electronic devices give grid operators far more flexible control over voltage, frequency, and power flow, essential for integrating distributed and intermittent renewable generation.

3. Clean energy conversion and green industry: high-efficiency converters minimize conversion losses, and variable-frequency drives cut energy waste in industrial motors, pumps, and compressors.
4. Electric/hybrid vehicles and electric aircraft: battery management systems, chargers, and traction inverters built on power electronics make electrified transport viable.

Data

Global electricity demand has grown steadily over the past three decades, and the composition of global generation is shifting toward renewable sources, which is only possible because of power-electronic conversion equipment. The figures below, compiled from IEA data on Bangladesh, illustrate representative national-level trends in electricity consumption, emissions, and cross-border trade that are shaped by this transition.

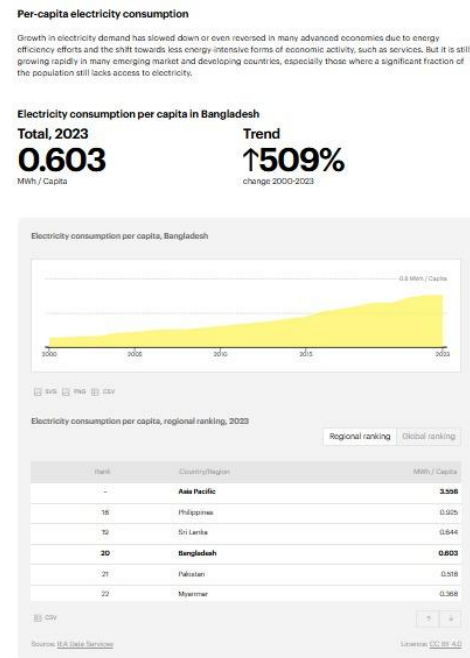


Figure 1 (left): Final consumption of electricity by sector, Bangladesh, 2023. Figure 2 (right): Per-capita electricity consumption and regional ranking, Bangladesh, 2023. (Source: IEA [2])

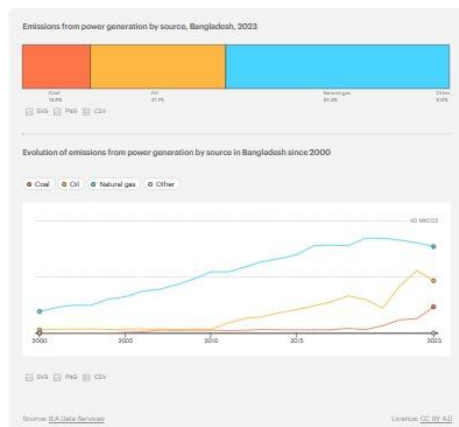
CO2 emissions from power generation

Power generation, which includes electricity and heat, is one of the largest sources of CO2 emissions globally, primarily from the burning of fossil fuels like coal and natural gas in thermal power plants.

Share of power in emissions

56%

of total energy-related CO2 emissions, 2023



Electricity imports and exports

Unlike other energy commodities such as coal, oil and natural gas, electricity trade between countries is relatively limited as it is more technically complex and requires a direct cross border interconnection. Such connections can help to balance out supply and demand across regions, which will be increasingly important as variable renewables like solar and wind make up a larger share of electricity generation.

Net electricity imports

10.7%

of 2023 total electricity final consumption

Trend in electricity imports

↑360%

change 2014-2023

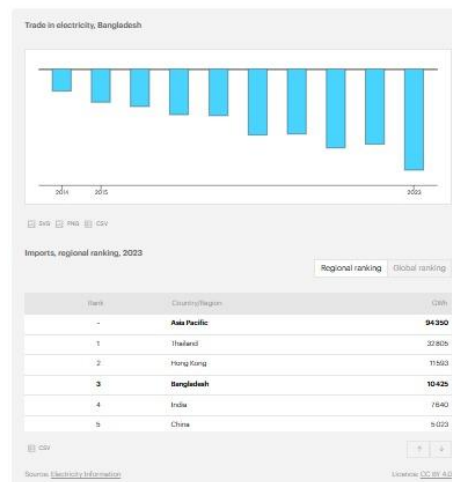
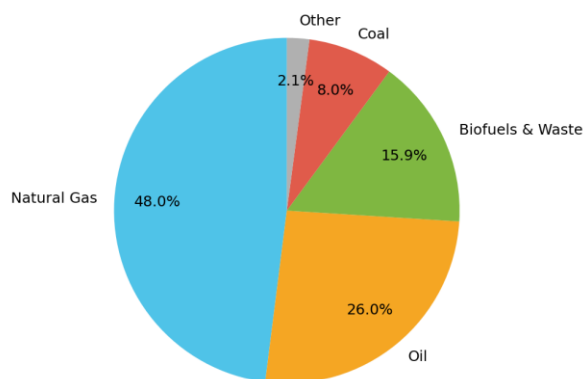


Figure 3 (left): CO2 emissions from power generation, Bangladesh, 2023. Figure 4 (right): Electricity imports and exports, Bangladesh, 2014-2023. (Source: IEA [2])

Bangladesh Energy System

Indicator	Value
Total energy supply – Natural gas	48%
Total energy supply – Oil and oil products	26%
Total energy supply – Biofuels and waste	15.9%
Total energy supply – Coal	8%
Energy-related CO2 emissions	106 Mt CO2 (0.31% of global emissions)
Change in emissions since 2000	+429%
Electricity consumption per capita	0.603 MWh / capita (+509% since 2000)
Renewables share of power generation	1.904%
Net electricity imports	10.7% of final consumption (+360% since 2014)



Bangladesh total energy supply mix, 2023 (Source: IEA [3]).

Future Prospect

The role of power electronics in the energy sector is expected to expand considerably over the coming decade. Wide-bandgap semiconductors such as silicon carbide (SiC) and gallium nitride (GaN) are beginning to replace conventional silicon devices in high-power converters, offering lower switching losses and higher efficiency. Grid-forming inverters are emerging as a key enabler for power systems with very high shares of renewable generation. Other significant trends include battery energy storage paired with power-electronic converters, electric-vehicle charging infrastructure, DC microgrids for rural electrification, and power electronics in green hydrogen production through electrolysis.

Bangladesh Perspective

Bangladesh has made substantial progress in electricity access, with coverage rising from about 20% of the population in 2000 to roughly 85% today, but the generation mix remains dominated by fossil fuels, principally natural gas. Figure 5 summarises the country's overall energy system as compiled by the IEA.

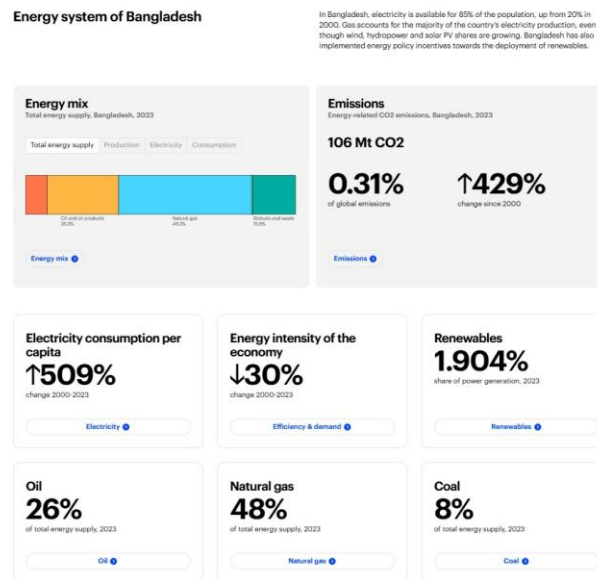


Figure 5: Energy system of Bangladesh (Source: IEA [2])

Within the electricity mix, renewable sources still account for a small share of total generation, though solar photovoltaic now dominates the renewable sub-mix, as shown in Figure 6. Figure 7 gives the corresponding renewable-energy installed capacity by technology reported by BPDB/SREDA.

Solar PV
66%
 of total generation from renewables

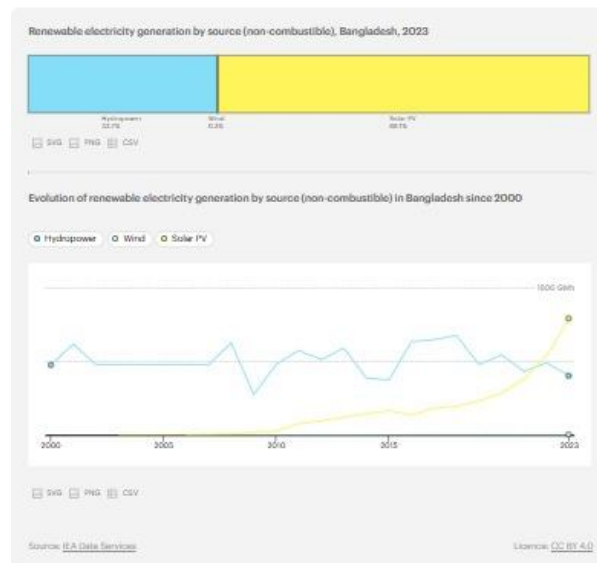


Figure 6: Solar PV share of renewable electricity generation, Bangladesh (Source: IEA [2]).

At the national grid level, the Bangladesh Power Development Board (BPDB) reported a total installed generation capacity of 29,593 MW as of August 2026, dominated by gas, coal and heavy fuel oil (HFO), as shown in Figure 8.

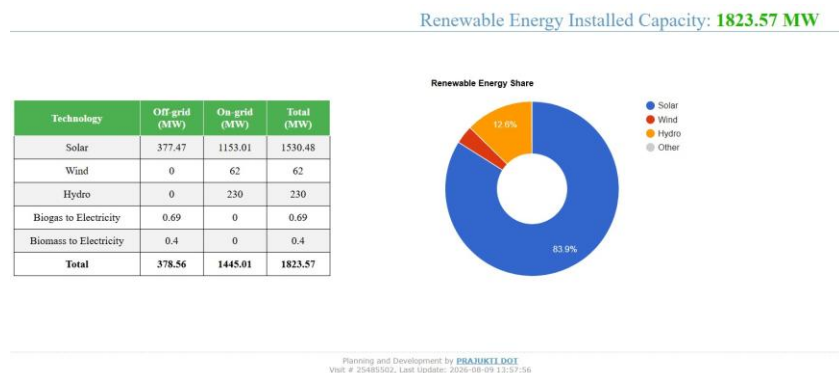


Figure 7: Renewable energy installed capacity by technology, Bangladesh (Source: BPDB/SREDA [3]).

Installed Capacity of BPDB Power Plants as on August 2026		
Fuel Type	Capacity(Unit)	Total(%)
Gas	12472 MW	42.15 %
HFO	5641 MW	19.06 %
HSD	768 MW	2.60 %
Coal	8423 MW	28.46 %
Imported	1160 MW	3.92 %
Solar	797 MW	2.69 %
Hydro	230 MW	0.78 %
Wind	62 MW	0.21 %
none	40 MW	0.14 %
Total	29593 MW	100.00 %

Figure 8: Installed capacity of BPDB power plants by fuel type, August 2026 (Source: BPDB [4]).

Installed capacity, however, overstates what the system can deliver: ageing units, water and fuel limitations, and technical constraints reduce the derated (dependable) capacity to 28,793 MW, as shown in Figure 9, a difference of 800 MW concentrated mainly in gas and coal units.

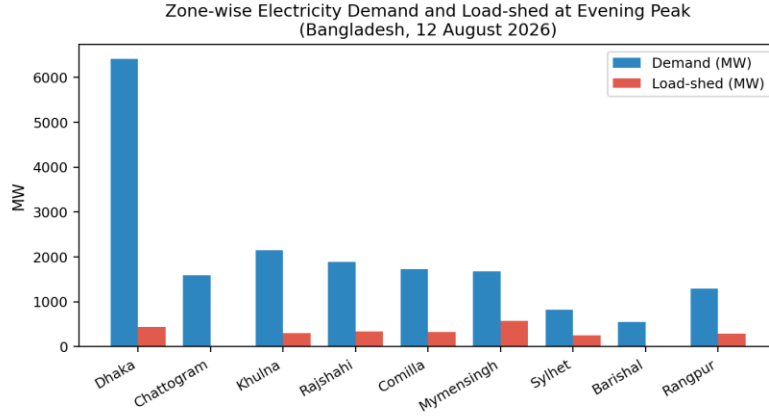
Derated Capacity of BPDB Power Plants as on August 2026		
Fuel Type	Capacity(Unit)	Total(%)
Gas	12154 MW	42.21 %
HFO	5637 MW	19.58 %
HSD	768 MW	2.67 %
Coal	7945 MW	27.59 %
Imported	1160 MW	4.03 %
Solar	797 MW	2.77 %
Hydro	230 MW	0.80 %
Wind	62 MW	0.22 %
none	40 MW	0.14 %
Total	28793 MW	100.00 %

Figure 9: Derated (dependable) capacity of BPDB power plants by fuel type, August 2026 (Source: BPDB [4]).

Despite this installed and derated capacity, actual peak generation is frequently constrained by fuel shortages and maintenance outages, producing a persistent gap between demand and supply. On 12 August 2026, the national evening-peak demand reached 18,106 MW against a delivered supply of only 15,568 MW, leaving a load-shed of 2,538 MW distributed unevenly across the country's nine power distribution zones, as shown in Figure 10.

Date			
12-08-2026		Search	
Sl	Zone Name	Demand (MW)	Load shed (MW)
1	Dhaka	6410	445
2	Chittagong	1591	5
3	Khulna	2148	303
4	Rajshahi	1894	337
5	Comilla	1724	324
6	Mymensingh	1673	575
7	Sylhet	825	256
8	Barisal	545	0
9	Rangpur	1296	293
Total		18106	2538

Figure 10: Zone-wise electricity demand and load-shed at evening peak, 12 August 2026 (Source: PGCB [5]).

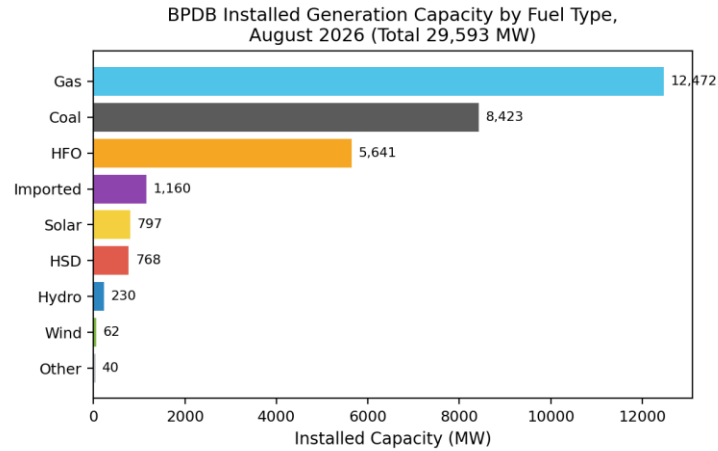


Zone-wise demand and load-shed at evening peak, 12 August 2026 (Source: PGCB [6]).

Table-level source-wise records published by BPDB break this generation down further, hour by hour, into gas, liquid fuel (HFO/HSD), coal, hydro, solar, and wind output, together with power imported from India (via the Bheramara HVDC link, the Tripura interconnector, and Adani Power) and Nepal. Figure 11 reproduces this hourly record for 15 August 2026, showing the evening peak at 21:00 reaching 16,190 MW, of which gas alone supplied 5,247 MW and coal 5,582 MW, while solar output fell to zero after sunset and Adani imports remained above 1,400 MW.

তারিখ	সময়	উৎপাদন(মেগাওয়াট)	গ্যাস	তরল জ্বালানী	কয়লা	হাইড্রো	সৌর	বায়ু	ভারত			নেপাল	মন্তব্য
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Figure 11: Hourly generation by fuel source and cross-border imports, 15 August 2026 (Source: BPDB [4]).

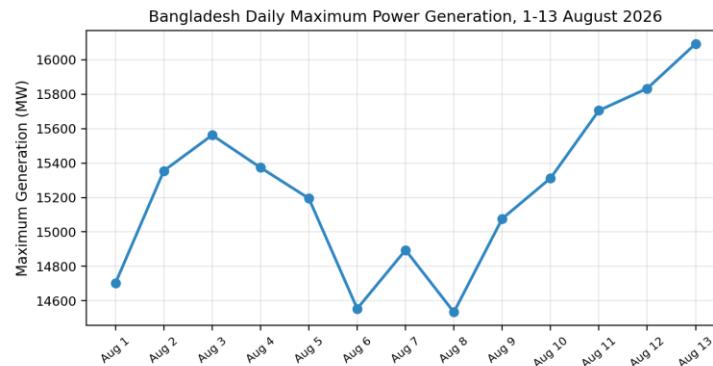


BPDB installed generation capacity by fuel type, August 2026 (Source: BPDB [4]).

Figure 12 shows the day-to-day maximum generation recorded over a thirteen-day window in early August 2026, which fluctuated between about 14,500 MW and just over 16,000 MW as fuel availability and temperature varied, before the daily generation curve in Figure 13 shows how that maximum is distributed across the day.

Date	Maximum Generation
2026-08-01	14702.08 MW
2026-08-02	15355.48 MW
2026-08-03	15562.48 MW
2026-08-04	15373.68 MW
2026-08-05	15194.98 MW
2026-08-06	14554.230000000001 MW
2026-08-07	14894.779999999999 MW
2026-08-08	14511.880000000001 MW
2026-08-09	15076.98 MW
2026-08-10	15112.68 MW
2026-08-11	15705.619999999999 MW
2026-08-12	15831.78 MW
2026-08-13	16091.28 MW

Figure 12: Daily maximum power generation, Bangladesh, 1-13 August 2026 (Source: PGCB [5]).



Daily maximum power generation, Bangladesh, 1-13 August 2026 (Source: PGCB [6]).

Figure 13 shows the corresponding daily generation curve recorded by PGCB's National Load Dispatch Centre, in which the total generation (red) tracks the sum of the East grid (blue) and West grid (green), peaking in the evening hours as residential and commercial demand rises.

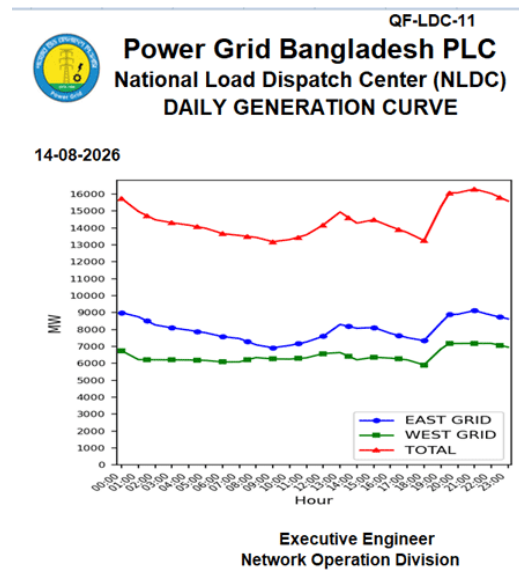
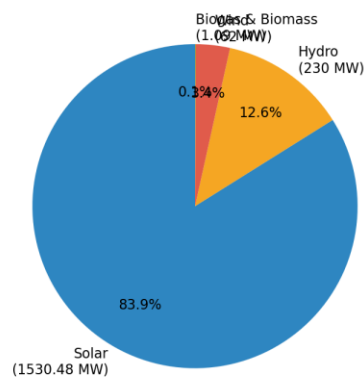


Figure 13: Daily generation curve, East grid vs West grid vs total, 14 August 2026 (Source: PGCB [5]).

Renewable Energy Installed Capacity, Bangladesh

Technology	Off-grid (MW)	On-grid (MW)	Total (MW)
Solar	377.47	1153.01	1530.48
Wind	0	62	62
Hydro	0	230	230
Biogas to Electricity	0.69	0	0.69
Biomasses to Electricity	0.4	0	0.4
Total	378.56	1445.01	1823.57

Renewable Energy Installed Capacity Share, Bangladesh
(Total 1,823.57 MW)



Renewable energy installed capacity share, Bangladesh (Source: BPDB/SREDA [5]).

These figures illustrate why power electronics is particularly important for Bangladesh. Efficient AC-DC-AC conversion in HVDC interconnectors (such as the Bheramara link with India) allows cross-border power imports to help cover shortfalls; power-electronic inverters are essential to the growing installed solar capacity already feeding the grid; and modern variable-frequency drives in industrial and irrigation loads can materially reduce the demand that must be met from scarce gas-fired capacity. As Bangladesh's renewable energy targets and planned solar parks come online, and as the national grid modernises toward smart-grid operation, power electronic converters will play an increasingly central role in closing the demand-supply gap sustainably rather than through further fossil-fuel capacity additions.

Conclusion

Power electronics underpins nearly every pathway toward a more sustainable energy system, from grid-scale renewable integration and smart-grid control to electric transport and industrial efficiency. Bangladesh-specific data show that demand continues to rise, fossil fuels still dominate generation, and renewable energy, while growing, remains a modest share of the total mix. Where natural gas supplies almost half of total energy and evening-peak load-shedding still reaches over 2,500 MW, wider deployment of efficient power-electronic converters, together with continued investment in solar, wind, and grid-interconnection capacity, offers one of the most practical routes to closing the demand-supply gap sustainably. Continued research, policy support, and infrastructure investment will be essential for achieving both energy security and environmental sustainability in the years ahead.

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